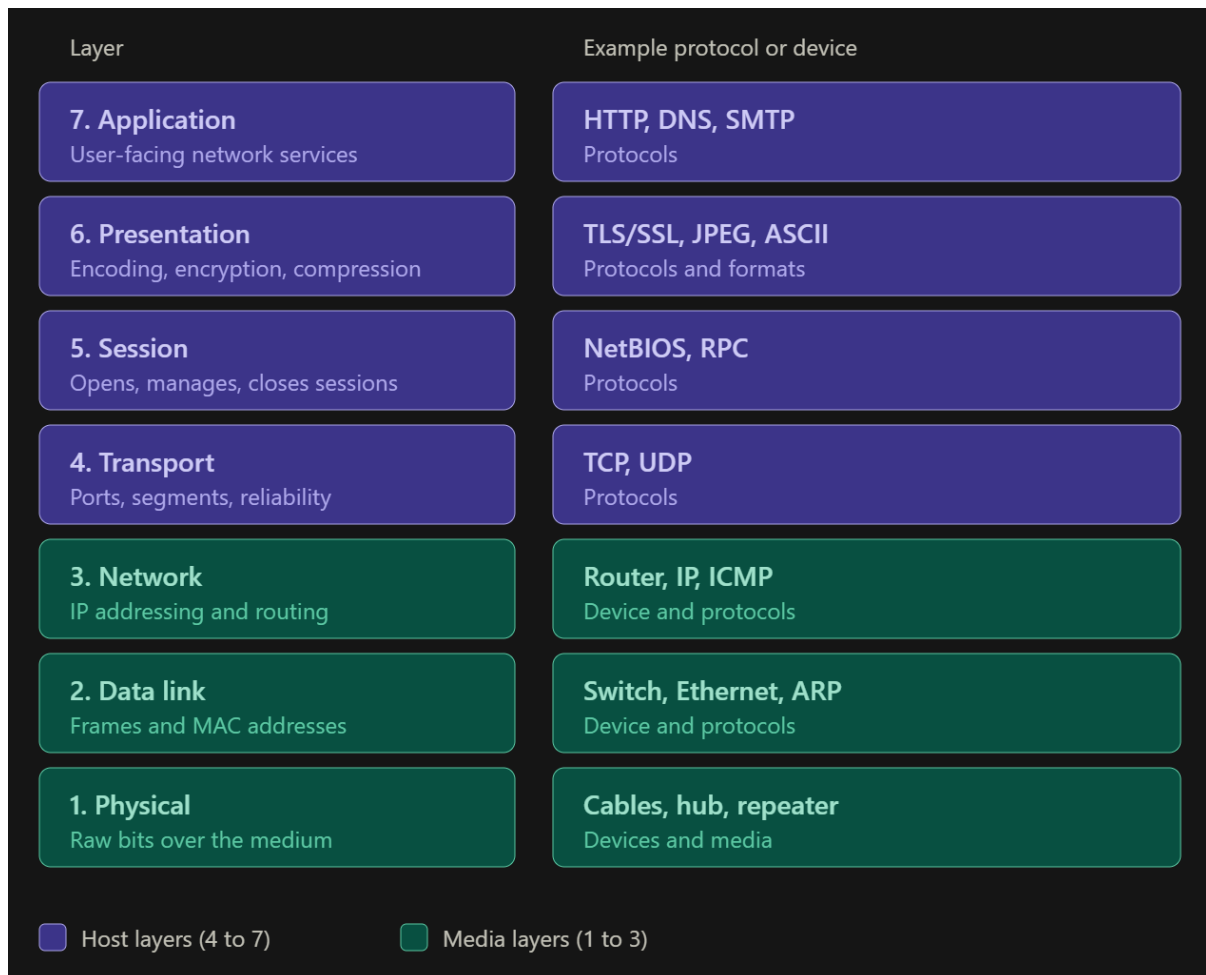


Osi Layer

1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

- **a) Encrypting data before sending:** Presentation Layer (Layer 6) – It handles data encryption, decryption, and formatting before transmission.
- **b) Routing packets across networks:** Network Layer (Layer 3) – It determines the best path and forwards packets between different networks.
- **c) Converting data to electrical signals:** Physical Layer (Layer 1) – It converts data into signals and transmits them through cables or wireless media.

- **d) Opening a web page in your browser:** Application Layer (Layer 7) – It provides network services directly to applications, such as web browsers using HTTP/HTTPS.
- **e) Detecting and correcting errors in frames:** Data Link Layer (Layer 2) – It manages frames and detects errors during communication between connected devices.

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Example: WhatsApp

When I send a message on **WhatsApp**, several OSI layers work together to deliver the data. At the **Application Layer (Layer 7)**, WhatsApp creates and manages the message being sent. The **Transport Layer (Layer 4)** helps transfer the data between my phone and the WhatsApp server using transport protocols. The **Network Layer (Layer 3)** uses IP addresses to route the data through different networks to reach the WhatsApp server. Finally, the lower layers handle the actual transmission through **Wi-Fi/mobile signals and network hardware**. When the server receives the message, the process works in reverse to deliver it to the receiver.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

If I can **ping the website's IP address successfully but cannot open the website**, the lower network layers are probably working, but the problem may be at the **Application Layer (Layer 7)** or **Transport Layer (Layer 4)**. The issue could be with the **HTTP/HTTPS service, web server, firewall, or TCP ports 80/443**. It could also involve **DNS (Layer 7)** if the website works by IP address but not by its domain name. Since ping works, basic IP connectivity through the Network Layer (Layer 3) is working.